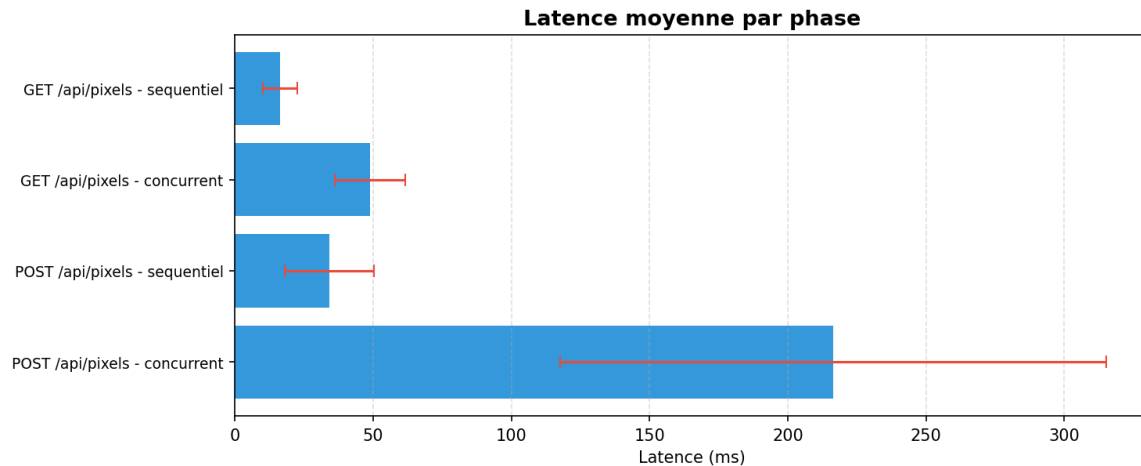
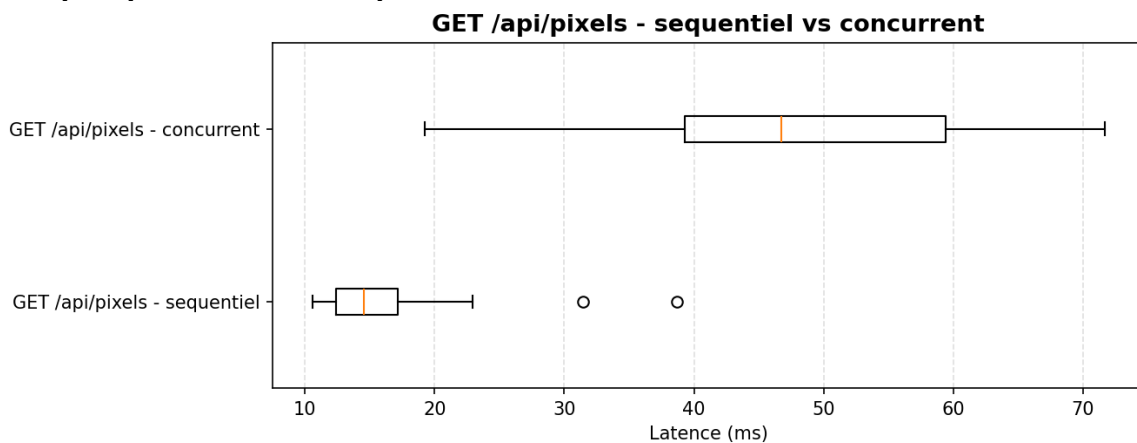


Pixel Wars — Rapport de benchmark V0

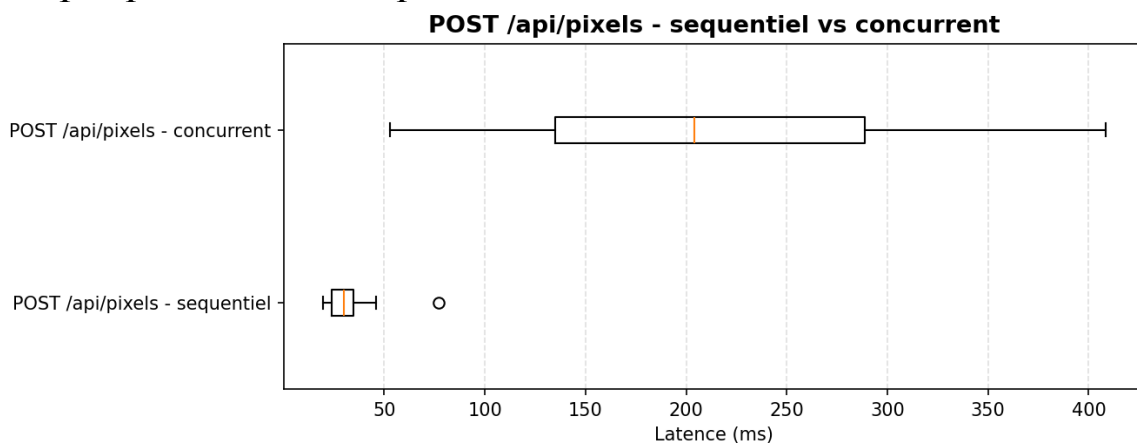
Graphique - Latence moyenne par phase



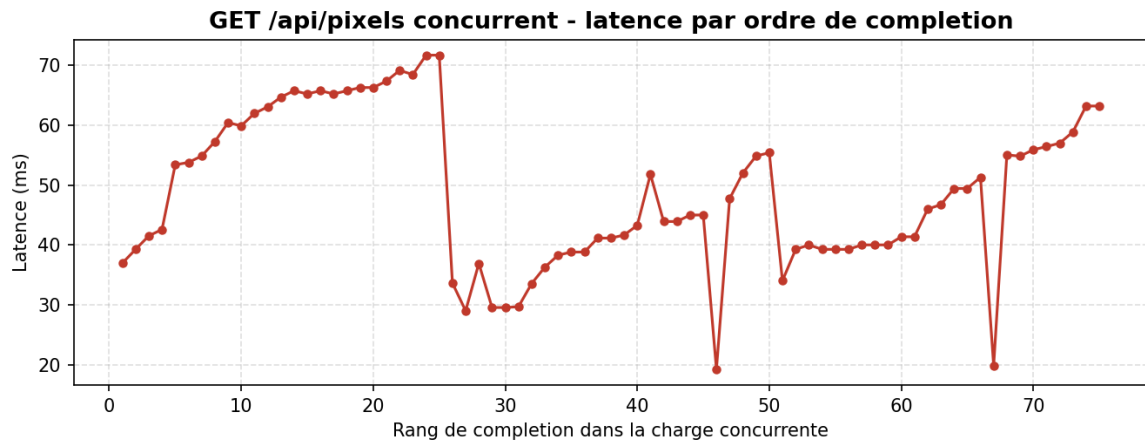
Graphique - GET : séquentiel vs concurrent



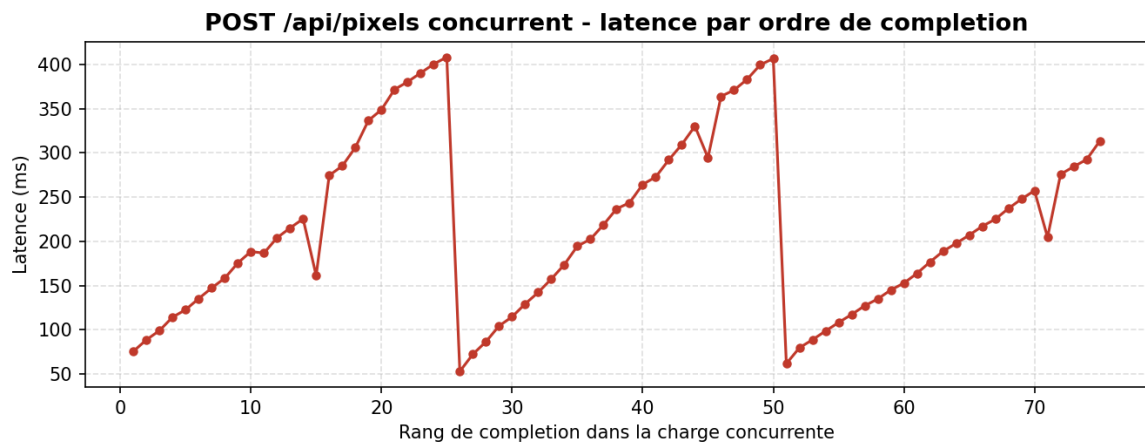
Graphique - POST : séquentiel vs concurrent



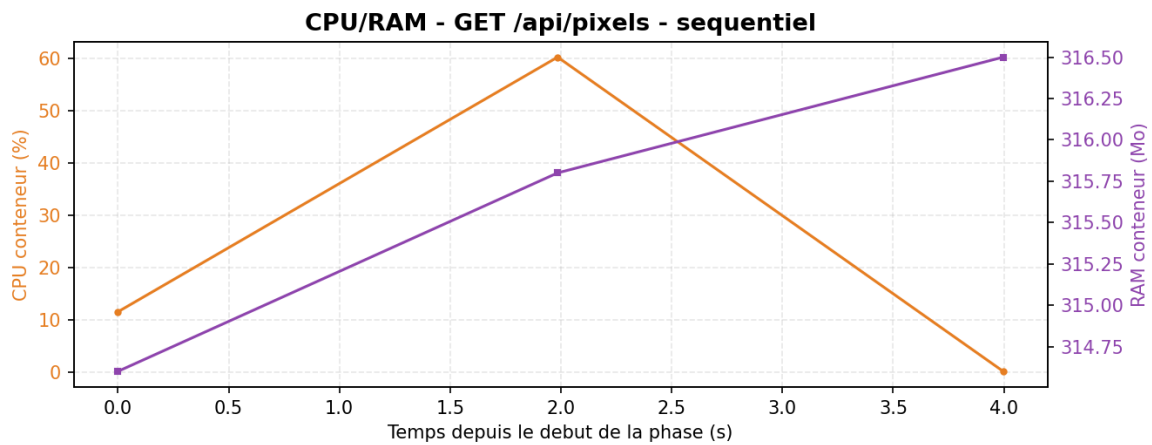
Graphique - GET : latence par ordre de complétion



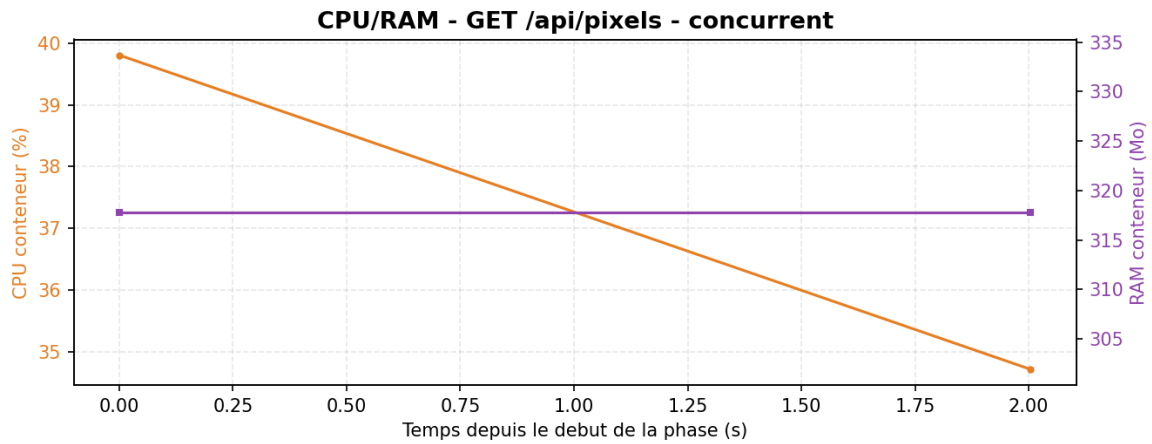
Graphique - POST : latence par ordre de compl  tion



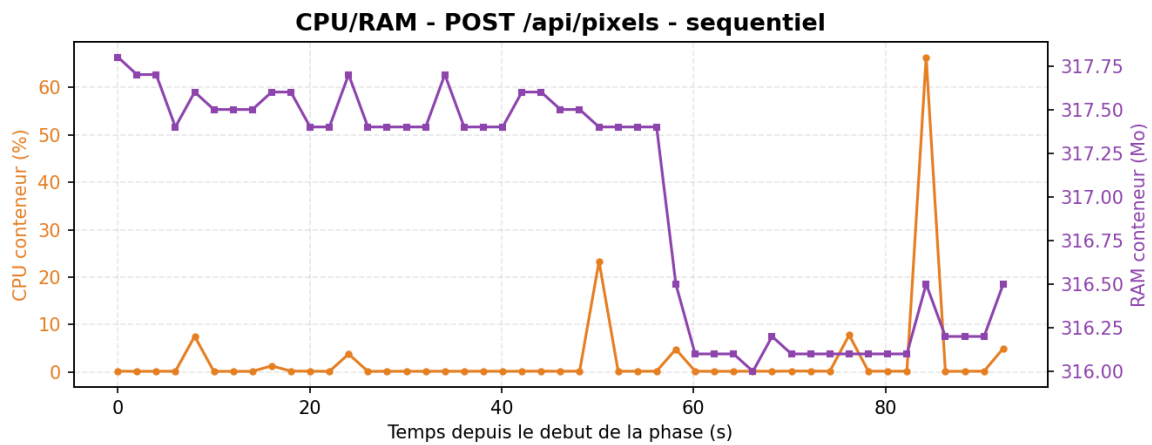
Graphique - Ressources CPU/RAM (GET s  quentiel)



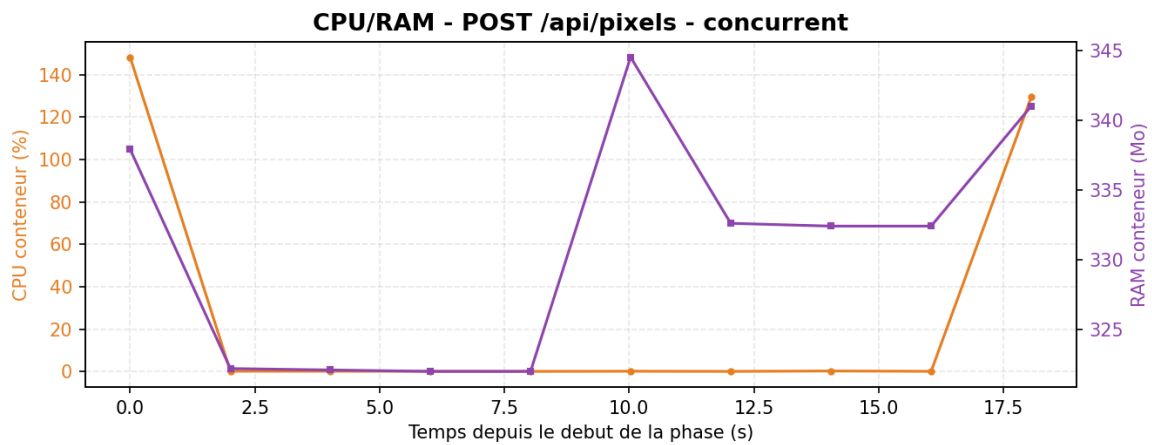
Graphique - Ressources CPU/RAM (GET concurrent)



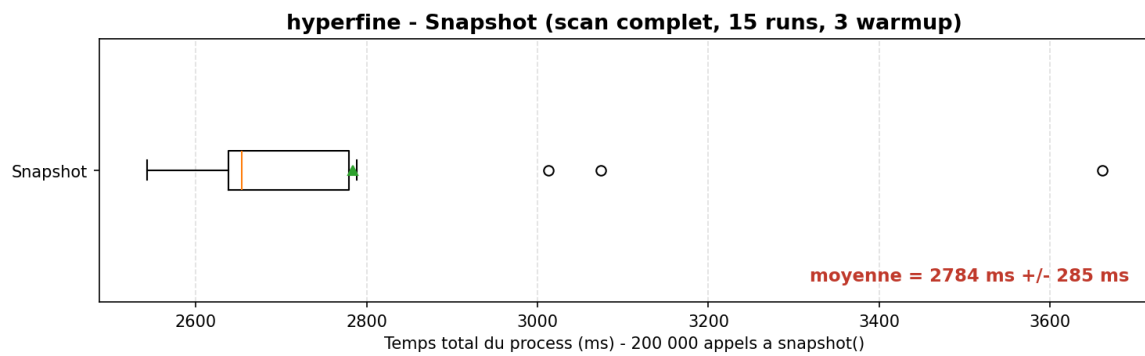
Graphique - Ressources CPU/RAM (POST séquentiel)



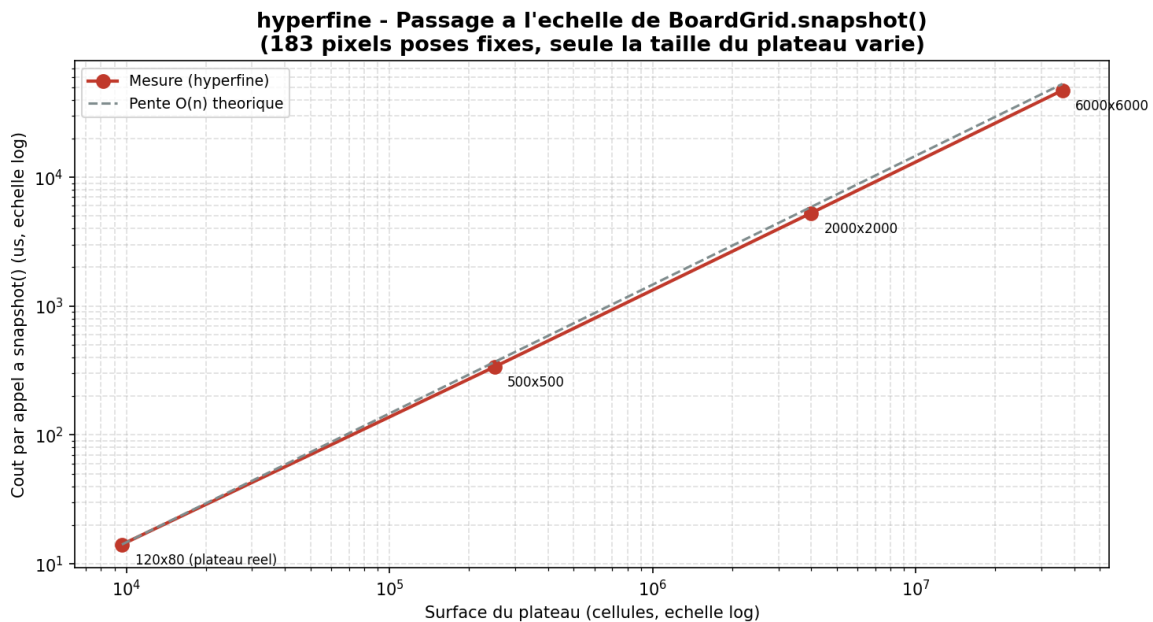
Graphique - Ressources CPU/RAM (POST concurrent)



Graphique - hyperfine : micro-benchmark de BoardGrid.snapshot()



Graphique - hyperfine : passage à l'échelle de BoardGrid.snapshot()



Graphique - Flamegraph : contention sur verrous (POST concurrent)

Phase	CPU moyen (%)	CPU max (%)	RAM moyenne (Mo)	RAM max (Mo)	N echantillons
GET /api/pixels - sequentiel	24.0	60.2	316	317	3
GET /api/pixels - concurrent	37.3	39.8	318	318	2
POST /api/pixels - sequentiel	2.7	66.3	317	318	47
POST /api/pixels - concurrent	27.9	148.3	331	345	10